

Overview

Problem Description

The competition, themed “Beyond the Black Box: Statistics for Trustworthy AI,” challenges participants to predict the probability of a 30-day hospital readmission using a synthetic healthcare dataset. The objective is not only to build an accurate predictive model, but also to demonstrate that the model is well-calibrated, robust, interpretable, and reliable across different groups and data conditions.

Start

3 days ago

Close

6 hours to go

Description

Challenge

You are part of an AI Validation and Statistical Assurance Team supporting a health system. Your task is to predict the probability that a patient will experience an unplanned readmission within 30 days.

Central question

Can you build an AI model we can trust-not merely one that tops the leaderboard?

Evaluation

Kaggle metric

The live Kaggle competition should use **Binary Log Loss**.

For predictions p_i and outcomes y_i :

Lower is better.

Why Log Loss? Because this challenge is about trustworthy probabilities, not only correct 0/1 classifications. Very confident wrong predictions are penalized heavily.

Required Kaggle submission

Exactly two columns:

```
patient_id, readmitted_30d
```

```
TE00001, 0.187421
```

```
TE00002, 0.731005
```

readmitted_30d must be a probability strictly between 0 and 1.

Final deliverables

Each team must submit:

- Final Kaggle prediction file
- Reproducible notebook/script that recreates the final predictions
- Two-page Model Trust Card using the provided template

Important warning

The test population may not be perfectly identical to the training population. Leaderboard chasing is therefore risky. Use statistical validation and sensitivity analysis.

Statistical evidence

Whenever you claim that one model is better than another, avoid relying on a single validation score.

Useful evidence includes:

- repeated cross
- validation
- bootstrap confidence intervals
- paired bootstrap comparisons
- calibration curves/intercept/slope
- Brier score
- subgroup confidence intervals
- sensitivity analyses

Prohibited

Attempting to infer hidden labels by leaderboard probing Manual label reconstruction Account sharing / submission collusion Using personally identifiable or real patient data Editing the test set to manufacture an advantage Any method that cannot be documented and reproduced

Allowed

Unless organizers announce otherwise:

- Open-source Python/R packages
- Pretrained generic software libraries
- Statistical / ML algorithms
- Internet documentation

- [AI coding assistants, only if the team discloses their use in the Trust Card](#)

What judges expect you to investigate

Your final solution should address:

- Predictive discrimination and probability quality
- Calibration
- Missing values and data quality
- Model stability / robustness
- Subgroup reliability
- Uncertainty and when the model should defer to a human
- Explainability
- Reproducibility
- Limitations and safe-use boundaries

Two-page Model Trust Card - template

Team

Team name:

Members:

Final Kaggle submission filename:

1. Model summary

Final model(s):

Key preprocessing:

Key hyperparameters:

Why this model was selected:

2. Validation design

Describe train/validation strategy.

Explain why it is appropriate.

Report uncertainty or variability across splits/resamples.

3. Performance

Report at minimum:

- Log Loss
- Brier score
- ROC-AUC
- Sensitivity / specificity at your chosen operating threshold

Do not report only the best single split.

4. Calibration

Include:

- calibration plot or summary
- calibration method, if used
- evidence before vs after calibration

5. Robustness

What might change between development and deployment populations?

What sensitivity tests did you run?

Which variables or modeling choices appeared unstable?

6. Subgroup reliability

At minimum examine:

- sex
- rurality
- age group
- hospital type

State group sizes. Report uncertainty where possible.

Do not interpret differences automatically as discrimination or causation.

7. Uncertainty / human referral

How did you identify predictions that should be treated cautiously?

If you abstain/refer the 10% most uncertain cases, what happens to performance on the remaining 90%?

8. Explainability

List the most influential predictors.

Provide one local explanation for:

- a high-risk prediction
- a low-risk prediction

Clearly distinguish **predictive association** from causation.

9. Failure modes

Give at least three concrete ways your model may fail.

10. Deployment recommendation

Choose one:

- Ready for limited prospective validation
- Requires additional model development
- Should not be deployed

Justify your choice in 100 words or fewer.

11. Reproducibility

Software / package versions:

Random seed(s):

Approximate training time:

AI-assistant use, if permitted:

12. One-sentence conclusion

Complete: **We trust this model only when...**

Competition Host

Dr. Sameera Viswakula

Prizes & Awards

Kudos

Does not award Points or Medals

Participation

28 Entrants

22 Participants

15 Teams

48 Submissions

Tags

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